



Detection of artificial intelligence-generated essays for academic assessment integrity using large language models

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ABSTRACT

Across various fields of human life, the lightning adoption of generative artificial intelligence (AI) tools is driven by their ease of use and ability to produce high-quality outputs, transforming communication and productivity. However, the growing popularity and influence of models such as ChatGPT have raised concerns in education regarding academic integrity, as AI-generated content challenges the authenticity of student work and complicates traditional assessment practices. Existing plagiarism detection tools, such as Turnitin and GPTZero, attempt to identify AI-generated content; however, their reliability remains limited, as most struggle to accurately differentiate between human and AI essays. To address existing tools limitations, we propose a novel detection model leveraging three large language models (LLMs): Generative Pre-trained Transformer 2 (GPT-2), Robustly Optimized BERT Pretraining Approach (RoBERTa), and Bidirectional and Auto-Regressive Transformers (BART). We first fine-tuned each model on two large-scale datasets to evaluate their effectiveness in distinguishing between human and AI-generated essays. To further enhance the performance, we applied both manual and automated hyperparameter optimization techniques, including Random Search, Grid Search, and Bayesian Optimization. Building on these experiments, we developed our BART-CNN model, which incorporates the best-performing BART configuration with an additional convolutional head classifier. Our BART-CNN model achieved impressive Macro F1-scores of 99.78% and 98.10% on the Kaggle and Hugging Face datasets, respectively, and demonstrated significant performance gains over baseline methods in cross-domain validation. Our study offers a critical advancement in AI plagiarism detection, helping to uphold academic standards and the assessment quality in an evolving AI landscape.

1. Introduction

In the contemporary era, the revolutionary advancement of artificial intelligence (AI) has profoundly impacted various sectors, including education (Kumar, Eaton, Mindzak, & Morrison, 2024; Mergel, Dickinson, Stenvall, & Gasco, 2023; Truby, Brown, & Dahdal, 2020; Wilson & van der Velden, 2022). In effect, AI has been widely adopted in educational institutions, evolving from basic computer technologies to advanced systems like humanoid robots and chatbots (Aleedy, Atwell, & Meshoul, 2022; Dogan, Goru Dogan, & Bozkurt, 2023; Labadze, Grigolia, & Machaidze, 2023; Xu & Ouyang, 2022). These AI systems have enhanced many administrative tasks, such as grading, and improved teaching quality by personalizing curriculum and content to meet individual student needs (Chen, Chen, & Lin, 2020; Malik et al., 2023). In general, AI applications in education focus on four key areas: intelligent tutoring systems, personalized learning, automated assessment, and teacher-student collaboration (Chen, Zhu, & Díaz del Castillo H., 2023; Kamalov, Santandreu Calonge, & Gurrib, 2023). First, the

intelligent tutoring systems employ adaptive algorithms, like reinforcement learning, as well as machine learning models and Natural Language Processing (NLP) techniques to provide customized feedback and create dynamic models of student learning. Second, through adaptive learning systems, AI personalized learning accommodates unique student demands, analyzes student performance, adjusts the curriculum's pace and complexity, and offers tailored resources. Furthermore, the automated assessment streamlines grading, delivering fast and personalized feedback, and boosting the student's engagement and outcomes. Finally, AI enhances teacher-student collaboration by offering real-time insights, allowing educators to adjust their teaching strategies based on students' strengths and weaknesses. Together, these AI applications are transforming education, making it more personalized, efficient, and collaborative, while expanding global access to quality learning.

While AI offers immense potential to transform education, it also brings significant challenges (Crook & Nixon, 2021; Uzun, 2023). Data privacy and security are key concerns, as students' sensitive information may be vulnerable to breaches or misuse. Another concern is bias and

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discrimination in AI systems, which can perpetuate existing inequalities by disadvantaging certain groups of students based on race, gender, or cultural background (Kamalov et al., 2023). To address these challenges, a strong ethical framework for AI in education is essential, encompassing principles such as responsibility, transparency, safety, accountability, inclusiveness, non-discrimination, and respect for user autonomy (Kwon, 2023). For instance, responsibility requires that educators and developers take proactive steps to ensure that AI systems are used ethically, serving the intended educational objectives and avoiding misuse. Transparency involves clear communication about how data is used, how algorithms work, and what regulations apply, allowing for informed consent and the protection of data ownership. As well, safety emphasizes the need for rigorous testing and monitoring to confirm that AI tools operate reliably and do not introduce risks to students or educators. Besides, privacy and security protecting students information is crucial to safeguarding user data from breaches, maintaining trust in these systems. Effective governance of AI in education should also involve interdisciplinary perspectives, ensuring that diverse insights shape policy and practice. Likewise, accountability requires that clear responsibility for AI actions be assigned, with a commitment to reducing potential harms. Additionally, inclusiveness and non-discrimination aim an equitable access to AI tools and prevent biases to ensure that AI benefits all students. Finally, the respect for user autonomy needs a human-centered approach to AI design to ensure that these technologies enhance human abilities, preserving individual freedom of choice and control.

On the other hand, the emergence of sophisticated generative AI models, such as Chat Generative Pre-Trained Transformer (ChatGPT), Gemini AI, and Large Language model Meta AI (LLaMA), has garnered significant attention in education for their ability to enhance student learning (Chen et al., 2023; Jen & Salam, 2024). These AI models provide personalized feedback, facilitate the understanding of complex concepts, and support the development of writing skills (Montenegro-Rueda, Fernández-Cerero, Fernández-Batanero, & López-Meneses, 2023). Notably, ChatGPT encourages active participation and collaboration among students, making it a valuable tool for scientific writing and group projects (Lai, Cheung, & Chan, 2023). It has the remarkable ability to produce text that closely mirrors human writing, generating high-quality content that can be challenging to distinguish from student-created work. While this capability offers educational benefits, it should not replace traditional cognitive processes or other learning resources (Li, 2023). Educators must integrate it thoughtfully, considering risks such as data privacy, cultural differences, and ethical implications. Otherwise, generative models could pose a significant challenge to the authenticity of student submissions, raising concerns about fair assessment (Sweeney, 2023; Swiecki et al., 2022). In fact, addressing these challenges requires careful regulation, strong data protection measures, and the development of fair and transparent AI systems to ensure ethical and equitable use in education. As the prevalence of AI-generated content grows, it becomes increasingly critical to develop and implement effective methods for detecting such content. These measures are essential to maintain the credibility of academic assessments and upholding the standards of academic integrity in an AI-driven world (Sweeney, 2023). Therefore, we propose in this paper a system for detecting AI-generated essays using NLP methods and Large Language Models (LLMs). Specifically, we classify text content as either student-essay or AI-generated using four models: Generative Pre-trained Transformer 2 (GPT-2), Robustly Optimized BERT Pretraining Approach (RoBERTa), Bidirectional and Auto-Regressive Transformers (BART), and a hybrid BART-CNN model to validate consistent performance across various dataset types. The main contributions of our paper are as follows:

- We propose a system for classifying text as AI-generated or human-written essays using NLP techniques and LLMs.
- We conduct a comparative analysis of the classification performance of our fine-tuned LLMs.

- We examine the effectiveness of automated hyperparameter optimization versus manual tuning.
- Our best-performing BART-CNN model achieved superior accuracy, outperforming the baseline methods across diverse datasets.

The remainder of the paper is structured as follows. Section 2 reviews related work in the field. In Section 3, we present our methodology, detailing the dataset preparation, model fine-tuning and optimization, as well as the baseline models used for comparison. Section 4 reports and analyzes the experimental results, highlighting their performances relative to the baseline models. In Section 5, we propose potential use case scenarios for applying our best classification model in real-world contexts. Section 6 addresses ethical considerations and outlines practical safeguards associated with the deployment of such systems. Finally, Section 7 concludes the paper and outlines directions for future research.

2. Related work

Plagiarism remains a significant concern in education, challenging the integrity of academic work and necessitating effective detection methods (Moorhouse, Yeo, & Wan, 2023; Uzun, 2023). Traditional plagiarism undermines this integrity by allowing students to present others' work as their own. This issue has become even more complex with the rise of AI-generated content. Unlike conventional plagiarism, AI-generated plagiarism involves creating seemingly original work using advanced language models, making detection significantly more challenging (Uzun, 2023). To address both types of plagiarism, various detection methods have been developed, employing techniques that range from lexical and semantic analysis to syntactic and grammar analysis, as well as machine learning and deep learning algorithms (Dalalah & Dalalah, 2023; El Mostafa & Benabbou, 2020; Elngar et al., 2021; Mishra, 2023). In this section, we review these methods and examine recent advancements in detecting AI-generated text.

Actually, numerous plagiarism detection tools and methodologies have been carried out for both written essays and source code (Hoq et al., 2024; Kulkarni, Govilkar, & Amin, 2021; Ljubovic & Pajic, 2020). For instance, Dadkhah, Oermann, Hegedüs, Raman, and Dávid (2023) have proposed a machine learning approach using decision trees to detect fraudulent healthcare and medical research papers. They have analyzed datasets from the Web of Science and various journals to validate research articles, distinguishing legitimate peer-reviewed work from deceptive publications. Zouaoui and Rezeg (2022) have introduced the Multi-agents Indexing System, an innovative method that enhances plagiarism detection in Arabic documents by utilizing an Arabic ontology for semantic representation, demonstrating high performance on the AraPlagDet corpus dataset. El-Rashidy, Mohamed, El-Fishawy, and Shouman (2022) have addressed plagiarism detection across lexical, syntactic, and semantic dimensions by developing a comprehensive database capturing diverse text similarities for intelligent learning systems. They utilized convolutional neural networks (CNN) and long short-term memory (LSTM) models to detect plagiarism on the PAN 2013 and PAN 2014 benchmark datasets. Hambi and Benabbou (2020) have explored deep learning models including Siamese LSTM, Doc2vec, and CNN to identify various types of plagiarism, offering a system that supports dataset customization and document comparison with online sources. Rosu, Stoica, Popescu, and Mihaescu (2020) have employed attention mechanisms for enhanced text encoding and contextualization, outperforming traditional word-matching methods. Their comparative analysis of pre-trained models such as Global Vectors for Word Representation (GloVe), RoBERTa, and Bidirectional Encoder Representations from Transformers (BERT) showed that BERT-based systems are the most effective in detecting plagiarism. Additionally, several studies have focused on detecting plagiarism in programming source code. Ullah, Jabbar, and Mostarda (2021) proposed a multi-step approach to extract plagiarism in source codes. The process begins with cleaning and pre-processing raw code through tokenization, breaking the code into

meaningful tokens, and then applying a weighting scheme to estimate the significance of features within the codes. To reduce noise caused by redundant or irrelevant tokens in the source code, the authors performed a dimensionality reduction using Singular Value Decomposition (SVD), followed by Latent Semantic Analysis (LSA) to extract the core semantics of the code. Finally, the plagiarism detection is executed using cosine similarity, and topic modeling to group related features, validating the overall approach. Similarly, Viuginov, Grachev, and Filchenkov (2020) have introduced an innovative method for feature extraction from C++ source code to detect plagiarism in Association for Computing Machinery (ACM) contests. The authors employed an Extreme Gradient Boosting (XGBoost) model on a dataset containing 50 ACM problems and approximately 90,000 solutions. The XGBoost classification model extracted features from both the syntactic and lexical properties of the Abstract Syntax Tree (AST) and the disassembled source code, achieving, in a binary classification, an F1-score of 74.5 % on the test set. These researches illustrate advanced methodologies in source code plagiarism detection, emphasizing the importance of feature extraction and sophisticated classification techniques.

On the other hand, AI has evolved plagiarism concerns, extending beyond traditional human-authored text to encompass AI-generated essays. This shift highlights a pressing need for detection methods that can accurately differentiate between student-written work and AI-produced content. For instance, Yeadon, Inyang, Mizouri, Peach, and Testrow (2023) have demonstrated that AI language models like ChatGPT and Davinci-003 can produce high-quality Physics essays with the highest academic classification available in the UK. Compounding the issue, standard plagiarism detection tools such as Grammarly and Turnitin have reported low plagiarism scores for these AI-generated texts, underscoring the threat these technologies pose to the integrity of essay-based assessments in academic settings, particularly in Physics courses. Likewise, Ventayen (2023) has used the Ouriginal plagiarism detection tool to evaluate the similarity between ChatGPT content and published papers from Pangasinan State University. The findings revealed that ChatGPT's generated text closely resembles existing academic work, exceeding the university's acceptable similarity threshold and thereby raising serious concerns about the potential misuse of AI for academic dishonesty. Also, Chaka (2023) has compared the performance of five AI content detection tools including OpenAI Text Classifier, Copyleaks, GPTZero, Giant Language model Test Room, and Writer.com's AI Content Detector to identify AI-generated content from YouChat, ChatGPT, and Chatsonic. Among the five tools, Copyleaks performed well, particularly with texts translated into different languages, while GPTZero struggled with non-English texts, often misclassifying them as human-written. In general, the study underscored the current insufficiency of AI detectors as reliable tools for detecting AI-generated plagiarism in academic writing across different contexts and languages. Weber-Wulff et al. (2023) have assessed the effectiveness of 12 publicly available tools and two commercial systems, including Turnitin and Plagiarism-Check, in distinguishing between human-written and AI-generated text, particularly from ChatGPT. The study found that these tools are generally unreliable, misidentifying AI-generated content from human essays, especially when content obfuscation techniques are applied. Collectively, these studies highlight the limitations of current detection tools and emphasize the need for more sophisticated and accurate algorithms to effectively identify AI-generated plagiarism in academic essays.

To address this growing challenge, Ibrahim (2023) has evaluated two RoBERTa-based classifiers on a dataset of 240 human-written and AI-generated essays to help educators control AI-assisted plagiarism, particularly from ChatGPT. Using the GPT-2 ODD (Output Detector Demo) and Crossplag AI Content Detector, the study achieved 89.2 % accuracy with the GPT-2 Detector, which proved more effectiveness at identifying human-written texts. Corizzo and Leal-Arenas (2023) have developed one-class learning models to detect AI-generated essays, focusing on L2 English and L2 Spanish, where traditional supervised meth-

ods and linguistic feature-based approaches primarily target L1 English data. The authors used a linguistic feature-based approach incorporating repetitiveness, readability, parts of speech, text characteristics, and emotional semantics to differentiate human-written text. They achieved 95.53 % accuracy with the AutoEncoder model on the English dataset and 72.38 % accuracy using the One-Class Support Vector Machine (SVM) on the Spanish dataset, with readability emerging as a key linguistic feature. Sağlam, Hahner, Schmid, and Burger (2024) have proposed an approach for detecting plagiarism in modeling assignments that combines automated analysis techniques with human oversight to ensure the detection accuracy. The approach has been examined on real-world assignments, including those obfuscated by ChatGPT, and demonstrated a significantly higher detection rate and broader resilience compared to existing methods. Cingillioglu (2023) has employed an SVM algorithm to identify human-generated essays from AI-generated essays. The author has used an n-gram bag-of-words discrepancy model and obtained a high accuracy compared to AI detection tools like GPTZero and Copyleaks. Sable et al. (2024) have developed a machine learning model called Extra Tree classifier (ETC), trained and tested on the ChatGPT Paraphrase dataset to distinguish between human and AI content. The model achieved 80.1 % accuracy, outperforming several baseline models including Decision Tree, SVM, and Linear Regression. Hoq et al. (2024) have employed machine learning and AST-based deep learning models to distinguish ChatGPT-generated code in student submissions. They have examined the models on a public student program for computer science assignments over ChatGPT generated code for the same assignments, and achieved 90 % accuracy. Mitchell, Lee, Khazatsky, Manning, and Finn (2023) have introduced a novel method for detecting text generated by LLMs named DetectGPT, which is based on a specific mathematical pattern calculated by the model. The DetectGPT leverages this pattern to identify AI-generated content without requiring a separate model or specialized datasets. It works by slightly altering the text and comparing these variations to the original model's calculations. The authors' approach achieved 95 % accuracy in identifying fake news generated by GPT-NeoX, outperforming the 81 % accuracy obtained with the zero-shot method. Li et al. (2023b) have utilized 14,908 reflection essays from pharmaceutical science students, using nine different prompts from ChatGPT, to examine the writing capabilities of generative LLMs. The writing quality has been assessed by human evaluators, AI detectors, and a BERT classification model. The authors applied metrics such as returning to experience, attending to feelings, association, integration, validation, and appropriation, with scores ranging from 0 to 6 for each criterion. The findings revealed that ChatGPT could produce high-quality reflective responses, outperforming student reflections in several metrics. Additionally, the BERT-based classifier was significantly more accurate in identifying AI-generated content than both human evaluators and a general-purpose AI content detector. Yan et al. (2023) have developed two detectors for AI-generated essays using SVM and RoBERTa models. The authors have evaluated their models on a dataset of 4000 human-written essays and 4000 AI-generated essays produced by the GPT-3 text-davinci-002 model. They obtained accuracies of 96 % with SVM and 99.75 % using RoBERTa. Zeng et al. (2024) have introduced a novel approach named TriBERT to overcome the challenge of hybrid text containing both AI-generated and human-written content in ChatGPT text. TriBERT, a boundary detection-based model, first separates AI and human content during encoder training, then identifies boundaries by analyzing dissimilarities between adjacent sentence prototypes. The authors' model demonstrated strong performance in detecting hybrid-generated essays, outperforming other baseline methods, including the GPTZero tool. Prajapati et al. (2024) have employed diverse texts and generative models to train feature learning aimed at distinguishing between human and AI essays. The authors demonstrated that the Generative Textual Likelihood Ratio (GLTR) annotation scheme can significantly increase the accuracy of AI-generated text detection from 74 % to 99 % without requiring prior training.

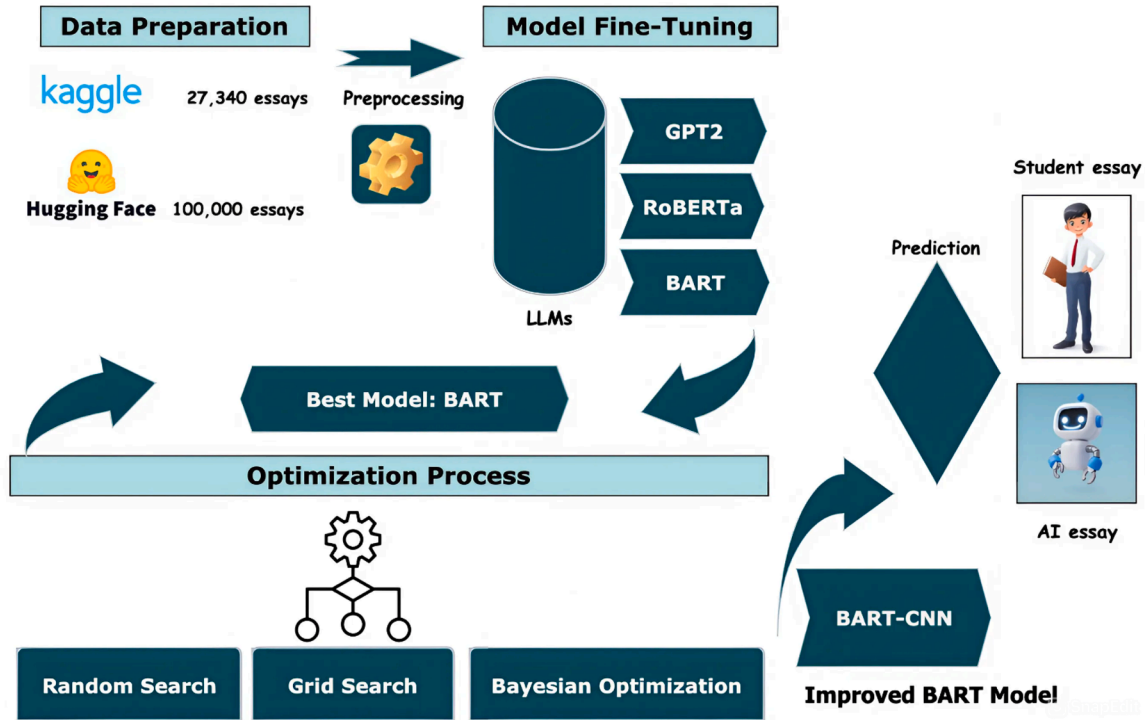


Fig. 1. The workflow of our proposed methodology for AI-generated essays detection.

Overall, the studies demonstrated the effectiveness of machine learning and LLMs in detecting AI-generated essays, outperforming many existing tools. However, notable gaps persist, including limited dataset sizes, model generalizability across different platforms, and reliance on specific baseline methods. Actually, most studies trained models on relatively small datasets generated specifically by ChatGPT, which can lead to overfitting and introduce bias, particularly when assessing essays with large volumes in real-world scenarios. Moreover, while some studies compare their top models to well known and commercialized plagiarism detection tools, these comparisons reveal limitations rather than demonstrating the full effectiveness of the proposed models. A more thorough analysis with algorithmic comparisons and benchmarking against state-of-the-art models is needed. Additionally, focusing solely on ChatGPT-generated content limits generalizability, as testing across a broader variety of AI sources and datasets would better demonstrate the proposed model robustness. Without such diversity, the models are likely to struggle with mixed-source or multi-origin essays, reducing effectiveness in real-world, and multi-source scenarios. In response to these gaps, our study adopts a methodology that fine-tunes three LLMs on two extensive datasets from Hugging Face and Kaggle. We then apply various hyperparameter optimization techniques and develop a hybrid deep learning and LLM-based classification model. Our method enhances the detection of AI- and human-written essays through an improved BART-CNN model, achieving performance that surpasses current state-of-the-art models. A detailed description of our methodology, along with the baseline models, is provided in the following section.

3. The Proposed methodology

Our proposed methodology aims to detect AI-generated essays by analyzing text content using LLMs. As shown in Fig. 1, we started by preparing two large datasets for training through pre-processing techniques and tokenization. These datasets include 27,340 essays from Kaggle and 100,000 essays from Hugging Face. Next, we fine-tuned three prominent LLMs: GPT-2, RoBERTa, and BART, specifically for our task to distinguish between human-written and AI-generated contents. Afterward, we conducted a comparative analysis of the models' performance

across the two datasets, which vary in size and sequence length. Through multiple fine-tuning experiments and parameter adjustments, we identified the optimal parameters of each LLM and designated BART as our robust baseline model for detecting AI-generated essays. Additionally, we explored the impact of different hyperparameter optimization methods, including random search (RS), grid search (GS), and Bayesian optimization (BO), on the accuracy of our best-performing models. This process provides another deeper analysis of the effectiveness of these optimization techniques in the context of AI-generating essay detection. Subsequently, we developed an enhanced BART-CNN model by integrating a CNN layer into the previous BART model. Finally, we evaluated its generalizability by applying it to all prepared datasets, including those from Kaggle, Hugging Face, and additional benchmark datasets, to assess its performance in classifying student-written and AI-generated essays.

As illustrated in Fig. 2, we used the encoder output of the pre-trained BART-base model, which generates contextualized token embeddings for each input sequence, as an input of a LayerNorm layer. Hence, we stabilized and normalized the hidden state distribution before being fed into a custom 1D convolutional layer. Then, the convolutional layer employed 128 filters with a kernel size of 3 and padding of 1 to capture local n-gram-like patterns across the token dimension. Next, we activated the convolutional outputs using ReLU function, and reduced the dimensionality using an adaptive max pooling layer, which ensures fixed-size output regardless of sequence length. This compact feature representation is then regularized using a dropout layer ($p = 0.3$) and passed to a fully connected linear layer with 2 output units, corresponding to the binary classification task (human vs. AI-generated). Practically, we applied a softmax activation function during inference to produce class probabilities, while the training is performed using cross-entropy loss on the raw logits, mixed-precision training with gradient scaling, and Adam optimizer. We also applied 3-fold stratified cross-validation to ensure robustness and reduce overfitting, given the class imbalance in some subsets of the data. All BART encoder layers remain fully trainable during fine-tuning, allowing the entire network – including the CNN feature extractor and classification head – to co-adapt and optimize for the downstream detection task. This hybrid architecture significantly improved the binary classification accuracy across both datasets, demonstrating

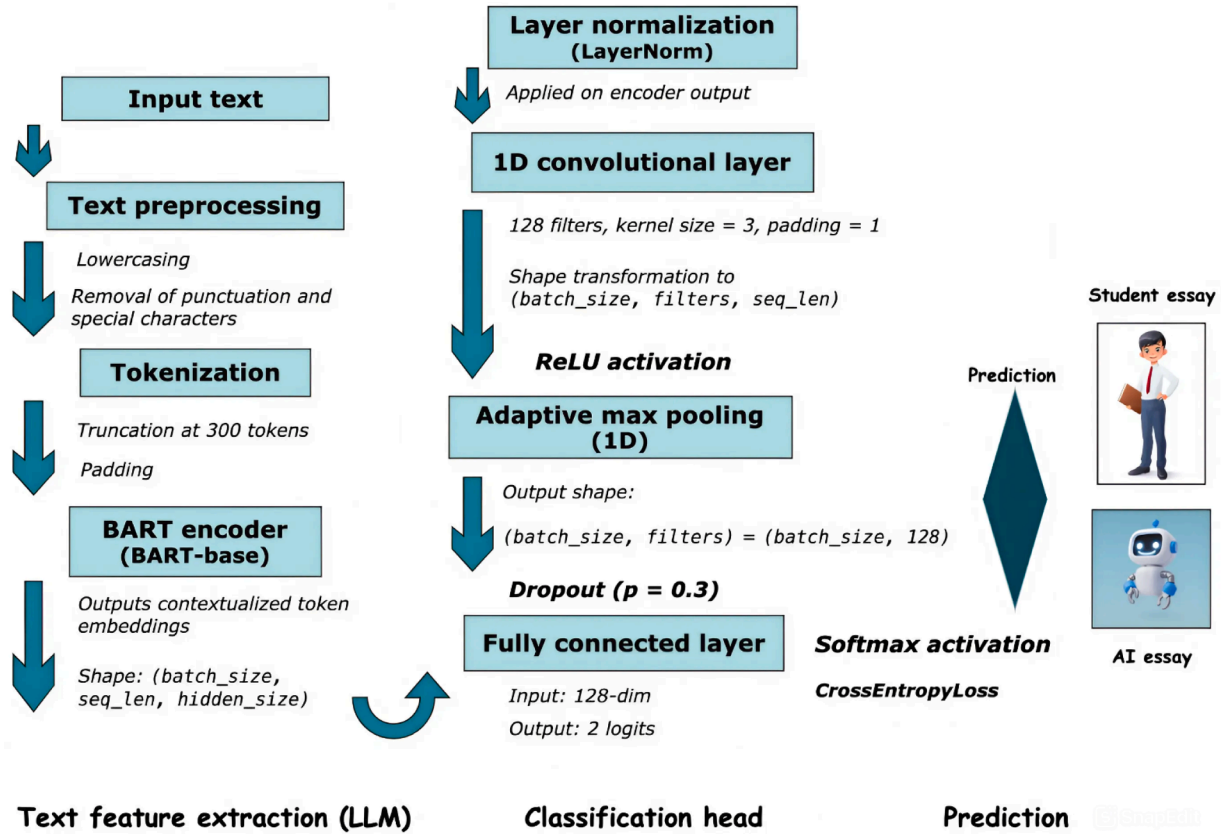


Fig. 2. The architecture of our proposed BART-CNN model for AI-generated essays detection.

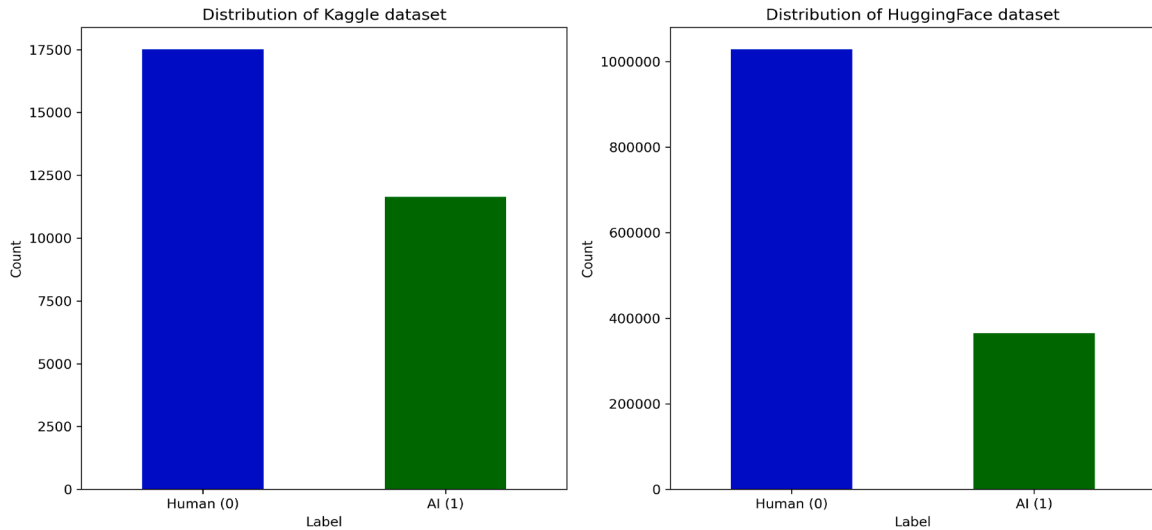


Fig. 3. Classes distribution in Kaggle and Hugging Face datasets.

the effectiveness of combining transformer-based language understanding with CNN-based feature extraction.

3.1. The datasets preparation

Dataset description

We conducted our experiments using two large-scale, publicly available datasets: the Kaggle AI Text Detection dataset (Kaggle, 2024) and the Hugging Face AI Text Detection Pile (Face, 2024). We selected these datasets for their diversity, scale, and relevance to the task of

distinguishing AI-generated text from human-written content. The Kaggle dataset is originally taken from the “LLM - Detect AI Generated Text” competition, organized by Vanderbilt University and The Learning Agency Lab (King, Baffour, Crossley, Holbrook, & Demkin, 2023). It comprises 27,340 labeled essay samples, with 17,508 written by students and 11,637 generated by various LLMs, including GPT-3, GPT-4, and Bard, as shown in Fig. 3. All essays were responses to one of seven prompts, two of which formed the public training set. The dataset is a CSV file format containing fields such as text (essay content), prompt_id, and generated (binary label: 0 for human-written, 1 for AI-generated). For model training, we extracted the text and generated

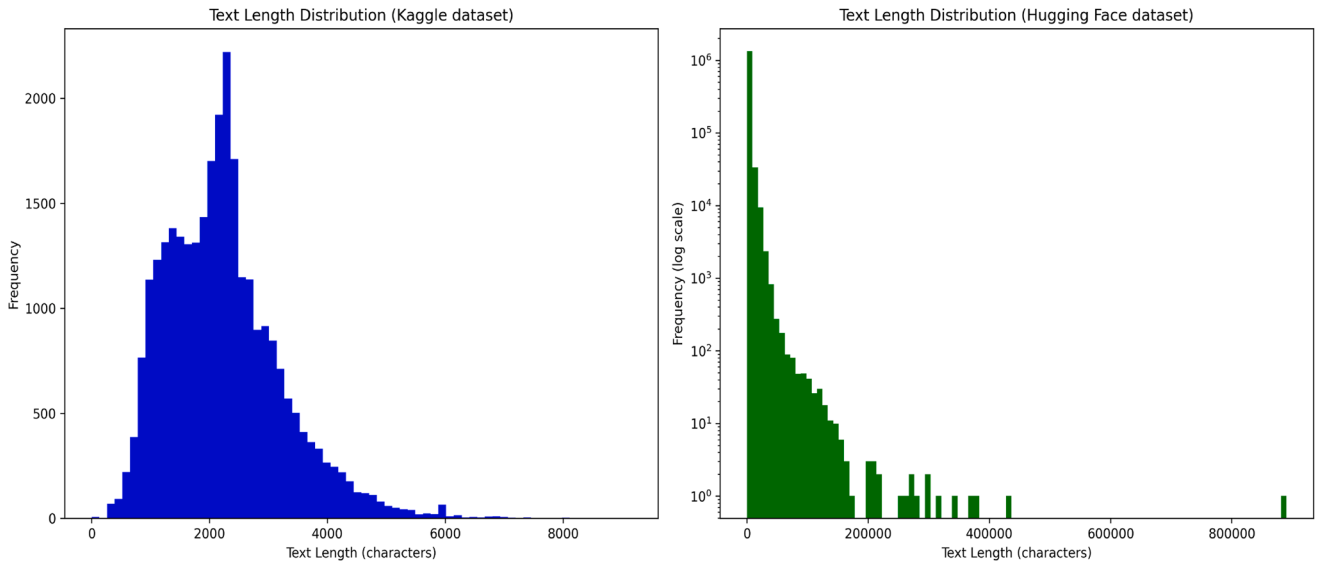


Fig. 4. Text length distribution in Kaggle and Hugging Face datasets.

fields to construct our Kaggle-based training dataset. In contrast, the Hugging Face dataset offers significantly broader coverage. It consists of approximately 1.39 million English text samples in Parquet format, including around 990,000 human-written texts and 340,000 AI-generated ones. Human texts were collected from sources such as Reddit Writing-Prompts (570k), OpenAI WebText (260k), HC3 human responses (58k), and IvyPanda essays, and the AI-generated texts were created by AI-models like GPT-2 (260k), GPT-3 (74k), GPT-J (44k), and ChatGPT (32.5k), derived from datasets including OpenAI GPT-2 outputs, pairwise instruct sets, and scraped Twitter conversations. Each sample includes a `text` field and a `source` label indicating whether the text is "human" or "ai-generated". We extracted essay contents from text fields and to standardize data in all used datasets, we converted source labels into binary values: 0 for human-written and 1 for AI-generated.

To address the class imbalance in both datasets, we applied class-weighted cross-entropy loss during training. Specifically, we assigned higher weights to the less frequent class (AI-generated essays) to counter the dominance of human-written essays, using a 3:1 ratio for the Kaggle dataset and inverse class frequency for the Hugging Face dataset. Throughout, we reported standard, unweighted evaluation metrics, accuracy and Macro F1-score, to ensure a balanced assessment without distortion by class proportions. Particularly for the Hugging Face dataset, we adopted another dual-validation strategy using our final BART-CNN model. First, we sampled 100,000 essays reflecting the original distribution (26.2 % AI, 73.8 % human) to evaluate the model under real-world imbalance. Then, we created a balanced subset of 100,000 essays (50 % AI, 50 % human) to test the model's robustness to controlled class distributions. Initially, we used stratified 70/30 train-validation splits to maintain class proportions across all LLMs training, and later transitioned to 3-fold stratified cross-validation to strengthen the reliability of our improved BART-CNN model. This comprehensive strategy, using weighted training and unweighted evaluation across both imbalanced and balanced scenarios, enabled us to validate our improved BART-CNN model performance consistently and confirm its resilience to varying class distributions.

Furthermore, Fig. 4 presents the text length distributions for both datasets, providing deeper insight into their structural differences. The Kaggle dataset exhibits a clear bell-shaped curve, with most texts ranging between 1000 and 4000 characters and peaking around 2000 characters. This consistent length distribution suggests that the dataset is relatively uniform, which can simplify model training and promote better generalization. In contrast, the Hugging Face dataset reveals a highly

skewed distribution. In fact, the majority of samples are short (under 10,000 characters), and the histogram (especially with the logarithmic y-axis) uncovers the presence of extreme outliers, with some texts exceeding 800,000 characters. These ultra-long samples significantly increase the dataset's variability and complexity. Consequently, models trained on this dataset may face challenges in processing such varied input lengths and might require strategies like truncation, hierarchical encoding, or segment-wise processing to handle them effectively.

In light of the differences in text length and complexity between the Kaggle and Hugging Face datasets, we implemented a consistent and robust preprocessing pipeline to address these challenges. Specifically, we standardized the text by converting all content to lowercase, removing digits, non-ASCII characters, and extraneous punctuation or noise. We further normalized whitespace and ensured consistent punctuation formatting. Furthermore, we used the pre-trained tokenizer associated with each respective LLM architecture to ensure compatibility with its downstream components. To manage computational efficiency and reduce input variability, we truncated all tokenized sequences to a fixed maximum length of 300 tokens and applied padding where necessary. This truncation has been particularly important for handling the extreme-length outliers found in the Hugging Face dataset, where some texts exceeded 800,000 characters. These steps ensured that the resulting sequences have been both semantically meaningful and model-friendly, reducing the risk of overfitting and improving generalization. Following preprocessing, we partitioned both datasets into a training set (70 %) and a validation set (30 %) for initial experimentation with LLMs. In contrast, we trained our final BART-CNN classifier using 3-fold stratified cross-validation to better leverage the full dataset while maintaining robust performance evaluation. Actually, we chose the 70/30 split to ensure a large and diverse training set capable of capturing underlying patterns, while also reserving a sufficiently large validation set for reliable performance assessment. Given the large scale of the datasets, this split enabled efficient training without compromising evaluation stability. Finally, the clean and standardized inputs prepared through this preprocessing pipeline have been fed into our classification models, enabling effective handling of variable-length texts and enhancing the overall training stability and performance.

While both datasets offer valuable diversity and scale, they present distinct limitations that must be considered when interpreting model performance and generalizability. The Kaggle dataset, though high-quality and grounded in academic writing, is limited in both prompt diversity and AI-generated content. Only two of the seven essay prompts

Table 1

Summary of datasets used for training and evaluation.

Dataset	Source	Size	Human:AI	Type	Models
Kaggle (2024)	Student essays	27,340	17,508:11,637	Essays	Mixed LLMs
Hugging Face (Face, 2024)	Web	100,000	74,000:26,000 50,000:50,000	Long-form text	GPT-2/3/J, ChatGPT
USE (Kyto, Axelsson, & Berglund, 2003)	CEFR essays	1,957	978:979	Essays	GPT-2
ChatGPT Para. (Vladimir & Maxim, 2023)	Quora, SQuAD, CNN	100,000	50,000:50,000	Questions/Sentences	GPT-3.5
Abstract Para. (Aleksandar, Miljana, & Staša, 2025)	Academic abstracts	304	152:152	Abstracts	GPT-4

are included in the public training set, and a relatively small proportion of the samples (approximately 42.5 %) are AI-generated. This imbalance may lead to model bias toward human-written text and reduced robustness when encountering essays outside the training prompt scope. On the other hand, the Hugging Face dataset provides broader coverage across domains and styles, incorporating both formal and informal texts from a wide range of sources. However, this breadth introduces variability in writing tone, structure, and intent. Moreover, many of the AI-generated samples are drawn from general-purpose or conversational contexts (e.g., Reddit, Twitter, WebText), which may not align with the stylistic and structural norms of academic essays. Furthermore, the AI texts originate from different models and timeframes, potentially introducing inconsistencies in writing quality and model-specific artifacts that affect the classification performance. Collectively, these limitations pose challenges to cross-domain generalization. To mitigate these issues, we trained our best-performing BART-CNN model on a diverse set of benchmark datasets encompassing academic writing, paraphrased content, and varying language proficiency levels. This approach aims to enhance model adaptability and robustness across a wider spectrum of real-world applications.

- **The Uppsala Student English Corpus (USE) dataset (Kyto et al., 2003):** It is a publicly available dataset containing 1,489 essays written by 440 Swedish university students learning English, spanning proficiency levels A1 to C1. Created at Uppsala University between 1999 and 2001 and published in August 2003, the dataset includes 1,221,265 words with an average essay length of 820 words. While Corizzo and Leal-Arenas (2023) used only 335 A2-level essays from this corpus, we significantly expanded its scope by incorporating 1,957 essays, where 978 human-authored texts covering the full Common European Framework of Reference for languages (CEFR) spectrum (A1-C1) and 979 AI-generated counterparts. First, we extracted human essays from structured XML documents, preserving metadata (e.g., <title>, <doc> tags), and applied preprocessing steps such as whitespace normalization, length filtering (> 50 tokens), and format standardization. For the synthetic component, we generated AI essays using GPT-2-medium via HuggingFace's text-generation pipeline. We conditioned each AI essay on the CEFR level and topic of its paired human essay, employing tailored prompt engineering. We generated the essays using a temperature setting of 0.7 and a maximum token length of 1,024, followed by post-processing steps such as sentence-boundary truncation and XML structure retention. We also implemented quality control measures to exclude outputs that did not meet structural or length criteria. This balanced, paired dataset enables fine-grained comparisons between human and AI writing across proficiency levels and provides a robust foundation for evaluating linguistic patterns using our BART-CNN model.
- **The ChatGPT Paraphrase (ChatGPT Para.) dataset (Vladimir & Maxim, 2023):** It is a publicly available Hugging Face dataset designed to support robust training and evaluation of paraphrase detection and generation models. The dataset includes approximately 420,000 samples, each consisting of an original sentence or question paired with five machine-generated paraphrases. The source material was drawn from three well-established corpora: 247,138 questions from the Quora Paraphrase Dataset, 91,983 passages from SQuAD 2.0, and 80,076 sentences from the CNN News dataset. The

dataset is categorized by text type (question or sentence) and original source (quora, squad_2, or cnn_news). All paraphrases were generated using OpenAI's gpt-3.5-turbo model, leveraging its strong generalization and rephrasing capabilities. Each input text was fed into the model with instructions to produce five semantically equivalent but lexically diverse variants. The final dataset includes three main columns: the original text, a list of five paraphrases, and the metadata fields category and source. To validate the generalizability of our best-performing BART-CNN model, we extracted 100,000 samples from the original dataset, maintaining an equal split (50 % human-written, 50 % AI-generated). For the AI portion, we randomly selected one of the five paraphrased questions and sentences, while the human portion comprised the original corresponding human-authored texts. This dataset offers a rich resource for studying paraphrase generation, evaluating semantic similarity models, and fine-tuning systems like our BART-CNN model for tasks requiring nuanced lexical variation.

- **The Abstract Paraphrasing (Abstract Para.) dataset (Aleksandar et al., 2025):** It consists of English-language academic abstracts paired with GPT-generated paraphrases, originally generated by Kartelj, Mladenović, and Vujičić Stanković (2025). The source texts have been obtained from the University of Belgrade and University of Singidunum's publicly available PhD thesis repositories. For our research, we focused exclusively on the English subset. In practice, we downloaded the cleaned (.txt) version of the data shared by the authors and converted it into a structured (.csv) suitable for our model training and evaluation. Then, we formatted each row and labeled them as either human-written (0) or GPT-generated (1), resulting in a total of 304 balanced abstract pairs (152 human, 152 AI) dataset. Actually, the authors generated the abstracts using the ChatGPT 4.0 API, employing a zero-temperature setting to ensure determinism and preserve semantic fidelity while allowing for syntactic variation. Subsequently, they aligned the human-written and AI-generated texts at the document level to present the abstracts as formal academic summaries. This dataset is particularly valuable due to its domain specificity, formal register, and longer text format, making it a strong complement to the other benchmark datasets with more conversational or general-purpose language.

Overall, Table 1 summarizes the datasets used in our paper, including both the training and benchmark datasets. In fact, the used datasets represent a rich and diverse collection of writing samples across domains (academic, conversational, narrative), styles (formal vs. informal), lengths (sentences to full essays), and language proficiency levels. While smaller curated sets like USE and Abstract Paraphrasing allow for controlled studies and fine-grained linguistic analysis, the large-scale Kaggle and Hugging Face datasets offer real-world, noisy, and heterogeneous data distributions ideal for robust model training. This combination ensures that our BART-CNN model is exposed to a wide range of stylistic, topical, and generative patterns.

3.2. Models fine-tuning

Feature extraction involves transforming raw text into useful representations that capture important key patterns to enable the AI-generated essay detection. The use of LLMs provides rich, contextual

features that can significantly improve the performance of our methodology. Specifically, LLMs generate contextualized embeddings that consider the entire surrounding text when creating features (Shanahan, 2024). This process is critical for tasks like human and AI-essay classification where context, fluency, and coherence are key. Besides, the LLMs are pre-trained on massive text corpora, which means they involve a vast understanding of language patterns, structure, and meaning (Shanahan, 2024). This allows them to produce rich, high-quality features from text without needing domain-specific engineering. For our binary classification task, we fine-tuned three LLMs with distinct architectural characteristics: GPT-2, RoBERTa, and BART. Each model is designed differently, which effectively provides rich feature identification methods, and insightful comparative analysis.

- **GPT-2** (Alec et al., 2019; Babaei & Giudici, 2024): is a transformer-based language model designed primarily for text generation tasks and it is pre-trained on large-scale text data. It has a decoder-only architecture and uses a Causal Language Modeling (CLM) objective, where the model learns to predict the next word in a sequence based solely on the preceding context. The model is trained using unsupervised learning, which makes it powerful at capturing long-range dependencies in text and adept at understanding complex sentence structures and flows present in our datasets' content. Additionally, its embeddings can capture deep contextual cues and detect unusual language patterns, which is particularly valuable for identifying AI-generated text.
- **RoBERTa** (Liu et al., 2019; Rodrawangpai & Daungjaiboon, 2022): is a large transformer-based model and an optimized version of BERT that improves the performance across a range of NLP tasks. It employs a bi-directional architecture and is pre-trained using the Masked Language Modeling (MLM) objective, where the model learns to predict randomly masked words within a sentence based on surrounding context. This bi-directional context understanding allows the model to capture contextual information from both directions in a sentence, enabling it to recognize subtle language patterns and nuanced details in our data. Additionally, its MLM design helps predict missing words, making it effective at identifying nuanced differences between human-written and AI-generated text in our essay classification task.
- **BART** (Lavecchia, 2024; Lewis et al., 2020): is an encoder-decoder transformer-based model that combines a BERT-like encoder with a GPT-like autoregressive decoder. BART is trained with a denoising objective, where text is corrupted (e.g., through permuting or masking) and the model learns to reconstruct the original text. This training approach equips the model with a deep understanding of text structure, coherence, and meaning, making it particularly well-suited for analyzing the complex structures of our essays data. Besides, its hybrid architecture allows BART to understand the structure of essays at a deeper level and make more accurate predictions about whether text is AI-generated or human-written.

Overall, these models provide comprehensive methodologies regarding the identification of AI-generated essay by analyzing both fine-grained linguistic details and larger text structures.

3.3. Optimization process

We fine-tuned the three LLMs (BART, GPT-2, RoBERTa) for our classification task using Kaggle environments with 16GB of RAM and access to the NVIDIA Tesla P100 GPU. To carry out this process, we utilized PyTorch, Hugging Face's Transformers library, Scikit-learn, NumPy, Pandas, and Optuna for hyperparameter tuning. For each model, we appended a fully connected linear layer with a softmax activation function (of size 2) to the pooled encoder output, applying mean pooling across token embeddings. During fine-tuning, all transformer layers were unfrozen and updated via backpropagation to ensure end-to-end learning. To encourage generalization and mitigate overfitting, we employed

Table 2

The hyperparameters utilized for the LLMs optimization process.

Model	Dataset	Learning rates	Batch size	Epochs	Optimization method
GPT-2	Kaggle	1e-5, 2e-5, 3e-5,	16	5	Manual tuning
RoBERTa		4e-5, 5e-5			
BART		5e-5	16,	5	
		1e-5, 5e-5, 1e-4	16, 32	3	Random search
GPT-2	Hugging Face	5e-5	16	5	Manual tuning
RoBERTa		5e-5	16	5	
BART		5e-5	16	5	
		5e-5	16, 32	3	Grid search
		5e-5 to 5e-4	16, 32	3	Bayesian optimization

mixed-precision training (AMP), dropout regularization ($p=0.3$), and the Adam optimizer. As illustrated in Table 2, we adjusted the learning rates, batch sizes, and epochs of our models on the two datasets, Kaggle and Hugging Face, using different optimization methods. First, we have employed manual experimentations starting with GPT-2 on the Kaggle dataset with a learning rate of 1e-5, batch size of 16, and 5 epochs. Through manual tuning, we gradually increased the learning rate from 1e-5 to 5e-5 to find the optimal setup. Next, we trained the RoBERTa and BART models using the best learning rate value of 5e-5 on the same dataset. Afterward, we applied the RS method to our best BART model, with a parameter space for learning rate, batch size, and epochs to compare with manual experiments and further refine its performance. On the Hugging Face dataset, we trained all the three models with a learning rate of 5e-5. For further performance and optimization analysis, we applied two additional methods to the best-performing BART model: GS and BO. The GS explored batch sizes while keeping the learning rate fixed at 5e-5 and epochs at 3. In contrast, BO explored learning rates in the range (5e-5, 5e-4) and batch sizes of either 16 or 32 to further improve accuracy, using the Optuna library to automate hyperparameter tuning. The optimization process employed a Bayesian search strategy, iteratively evaluating different hyperparameter configurations to maximize the F1-score on the validation set. For each trial, Optuna fine-tuned our BART model with gradient scaling (via GradScaler) for mixed-precision training, computed cross-entropy loss, and tracked validation metrics. To enhance search efficiency, we implemented early pruning to terminate underperforming trials. Through 20 iterations of Bayesian optimization using Optuna's Tree-structured Parzen Estimator (TPE), we identified the optimal hyperparameter configuration for our BART model. This selected configuration, when used for final training, demonstrated enhanced performance compared to both intermediate trial results and the other LLMs. Moreover, we visualized the training progress through accuracy and Macro F1-score curves to verify the model's convergence and robustness. This BO approach enabled computationally efficient hyperparameter exploration by strategically focusing on promising regions of the parameter space. Meanwhile, the optimization process was constrained by limited computational resources (maximum 100 GPU hours and limited memory), which influenced the choice of parameter ranges for the auto-optimization methods. Despite the large Hugging Face dataset size (100,000 entries), we carefully selected parameter ranges to balance experimentation with resource availability, ensuring an efficient and focused search for the best-performing model.

Practically, we chose the optimization methods based on dataset size and computational efficiency. The RS allows an efficient hyperparameter exploration without exhaustive computations, making it suitable for a moderately-sized dataset like Kaggle. Compared to manual tuning, the RS method improved the performance of our best BART model by effectively uncovering optimal parameter configurations. For the Hugging Face dataset, we compared manual tuning with the two other optimization methods GS and BO. Due to the lower performance achieved

on Hugging Face compared to Kaggle, we aimed to enhance the BART model's effectiveness on this dataset. Moreover, given the larger volume of the Hugging Face data and our available but limited training resources, we selected well-targeted optimization methods with fewer evaluations across a reduced hyperparameter space. By using GS and BO methods on the Hugging Face dataset, we balanced thorough exploration with computational efficiency, ultimately achieving optimal model performances.

Following the optimization of LLMs, the BART model consistently outperformed the GPT-2 and RoBERTa models. To further enhance its performance, we refined the BART architecture and leveraged the optimization process to configure it with optimal hyperparameters, establishing it as our final baseline validation model. Specifically, we integrated a CNN-based classification head into the BART encoder to take advantage of CNN's strength in capturing local textual features. Then, we trained our BART-CNN model on the Kaggle and Hugging Face datasets, using 3-fold cross-validation to ensure generalization and robustness across splits. In particular, we evaluated two versions of the large Hugging Face dataset: one with the original class distribution (26% AI, 74% Human), and the other with a rebalanced split (50% AI, 50% Human) to mitigate class bias and fully exploit the dataset's scale. Subsequently, we assessed the generalizability of our BART-CNN model on three benchmark datasets (USE, ChatGPT Paraphrase, and Abstract Paraphrase) to demonstrate its adaptability across different domains and writing styles. The BART-CNN model achieved strong performance across all datasets, confirming its robustness in distinguishing AI-generated from human-written content, regardless of domain complexity or dataset composition. In practice, we evaluated the performances of our models using unweighted accuracy and Macro F1-score, as defined in Eqs. (1) and (2) (Hinojosa Lee, Braet, & Springael, 2024; Takahashi, Yamamoto, Kuchiba, & Koyama, 2021). In this context, True Positives (TP) are correctly predicted instances of the AI class, False Positives (FP) are human-written texts misclassified as AI-generated, True Negatives (TN) are correctly identified human-written instances, and False Negatives (FN) are AI-generated texts incorrectly labeled as human-written (Asim, Rehman, & Shoaib, 2017). The Macro F1-score provides a balanced assessment by averaging F1-scores across both classes equally, regardless of class imbalance. These metrics offer a holistic view of the model's classification performance, emphasizing its ability to distinguish between AI- and human-generated content. Detailed results are presented in the following section.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$F1_{\text{macro}} = \frac{1}{2} \sum_{i \in \{\text{AI}, \text{Human}\}} \frac{2 \cdot \text{Precision}_i \cdot \text{Recall}_i}{\text{Precision}_i + \text{Recall}_i} \quad (2)$$

3.4. Baseline methods

The effectiveness of our proposed methods lies not only in their classification performance but also in their comparative analysis with other methods. Hence, we selected the baseline methods that share the same objective to clearly contextualize our paper. Several methods have been developed to detect AI-generated content, particularly from ChatGPT. Below is a descriptive overview of the models used in the baseline methods, including the datasets utilized and the results achieved.

- **The AutoEncoder** (Corizzo & Leal-Arenas, 2023; Li, Pei, & Li, 2023a): is a neural network designed for unsupervised learning that excels at capturing hidden patterns in unlabeled data. This model consists of an encoder that compresses an input data into a low-dimensional representation and a decoder that reconstructs the original input from its compressed form. The AutoEncoder is particularly effective for outlier detection, as it can evaluate reconstruction errors in new inputs to identify unusual patterns. In the context of

AI-generated essays, the model identifies different syntactic and stylistic features as outliers due to their higher reconstruction errors. Utilized by Corizzo and Leal-Arenas (Corizzo & Leal-Arenas, 2023), the AutoEncoder model achieved an F1-score value of 95.53% on the Uppsala Student English Corpus (USE). This dataset includes 335 argumentative essays written by Swedish university students at an A2 English proficiency level, along with an additional 335 essays generated by ChatGPT.

- **ETC** (Bhati & Rai, 2020; Sable et al., 2024): is a robust ensemble machine learning model that utilizes a tree-based approach to effectively distinguish between human and AI generated contents. This model employed the Extra Trees algorithm, which constructs multiple decision trees from random subsets of the training data and features, enhancing classification performance through aggregation. Utilized by Sable et al. (2024), ETC effectively classified AI-generated content surpassing several machine learning models, and achieving an accuracy of 80.1% on the ChatGPT Paraphrase dataset.
- **AdaBoostClassifier** (Kartelj et al., 2025): is a high-performing ensemble learning model that employs boosted decision trees to detect AI-generated text with notable accuracy. Kartelj et al. trained this model using Scikit-learn on TF-IDF vectorized character 4-grams (char4), allowing for fine-grained analysis of textual patterns. The classifier integrates nine iteratively tuned weak learners, with performance enhanced through adaptive boosting. It achieved a test accuracy of 97.2% on the Abstract Para. dataset, generated by the authors, and outperformed other baseline models such as XGBoost and Logistic Regression. Notably, it maintained a low false negative rate (FN = 49) on English academic abstracts. Although the total tuning process required substantial computational time (17,811 s), the model's ability to minimize false negatives makes it particularly suitable for high-stakes applications where missing AI-generated content carries significant risk.

Generally, the baseline methods have demonstrated effective performance in detecting AI-generated essays. Through neural network architectures and ensemble learning techniques, each approach tailored to enhance academic assessments and address the challenges posed by AI in education. However, these baseline models have been trained on single and smaller datasets, which limits their capacity to generalize across diverse types of content and varied linguistic styles. To overcome these limitations, we trained three LLMs: BART, GPT-2, and RoBERTa on two large and diverse datasets. Each LLM offers unique architectural advantages: BART's encoder-decoder structure, GPT-2's decoder-only approach, and RoBERTa's bidirectional modeling. All together, these LLMs enhance the detection of subtle linguistic variations, improving the identification of AI-generated essays. Additionally, we employed automated optimization techniques to fine-tune the hyperparameters, which significantly boosted our models' performance in the classification task. Furthermore, our developed BART-CNN model benefits from the large context and deep understanding architecture of BART alongside the consistent ability of CNN to extract local features. Overall, we contribute to the existing literature by introducing our proposed methodology and providing a comprehensive comparative analysis with the current state-of-the-art models. Further details and insights from our analyses will be presented in the following Results and Discussion section.

4. Results and discussion

The use of LLMs for detecting AI-generated essays yielded notable results on both Kaggle and Hugging Face datasets, as illustrated in Table 3. The table presents the implemented optimization methods, the hyperparameter values, as well as the accuracy and Macro F1-score of each experiment. Regarding the hyperparameters, 'Lr' refers to learning rate, 'Bs' to batch size, and 'Ep' to the number of epochs.

Effectively, starting with a learning rate of 1e-5, a batch size of 16, and 5 epochs generated a good initial result using GPT-2 on the

Table 3

The accuracy and Macro F1-score of the proposed LLMs on Kaggle and Hugging Face datasets.

Model	Optimization method	Hyper-params Lr, Bs, Ep	Kaggle dataset		Hugging Face dataset	
			Accuracy	F1-score	Accuracy	F1-score
GPT-2	Manual tuning	1e-5,16,5	94.37	94.37	–	–
		2e-5,16,5	96.53	95.59	–	–
		3e-5,16,5	97.34	96.63	–	–
		4e-5,16,5	97.64	97.01	–	–
		5e-5,16,5	98.03	97.50	88.99	79.33
RoBERTa	Manual tuning	5e-5,16,5	98.33	97.88	88.49	78.57
BART	Manual tuning	5e-5,16,5	98.55	98.16	91.26	83.84
	Random search	5e-5,32,3	99.78	99.72	–	–
	Grid search	5e-5,32,3	–	–	96.58	93.90
	Bayesian optimization	7e-5,32,3	–	–	97.2	94.54

Kaggle dataset. Next, the performance gradually improved as the learning rate increased, reaching 98.03% accuracy and a Macro F1-score value of 97.50% with the same GPT-2 model. This demonstrates that manual tuning can be an effective method for improving the model performance through multiple fine-tuning experiments. However, GPT-2 struggled to generalize its strong performance on the Hugging Face dataset, by achieving an accuracy of 88.99% and a Macro F1-score of 79.33%. On the other hand, RoBERTa showed a slight improvement over GPT-2 with manual tuning on the Kaggle dataset but performed slightly less on the Hugging Face dataset. This result suggests that the Hugging Face dataset has presented more challenges than Kaggle, at least for these manually tuned models. Afterward, the BART model outperformed both GPT-2 and RoBERTa on the Kaggle and Hugging Face datasets, with accuracies of 98.55% and 91.26%, respectively. With RS, BART improved its performance, achieving an accuracy of 99.78% and a Macro F1-score of 99.72%. In fact, RS successfully identified better hyperparameters than manual tuning, particularly with a larger batch size of 32 and 3 epochs. On the Hugging Face dataset, the GS method further enhanced the performance, obtaining high results and demonstrating its effectiveness in fine-tuning hyperparameters. Nevertheless, BO yielded the best results on Hugging Face dataset, reaching an accuracy of 97.2% and a Macro F1-score of 94.54%. With a learning rate of 7e-5, the BO method outperformed both manual and GS methods, demonstrating its strength in efficiently exploring the hyperparameter space.

Actually, the provided results can be attributed to different perspectives:

- **Dataset size and complexity:** The size, length, and complexity of the dataset heavily influence the performance of the models. On the smaller Kaggle dataset (27,340 sentences), all models performed exceptionally well, with only minor differences. Additionally, the uniformity in text lengths made it easier for the models to process. As mentioned in the Dataset Section (3.1), the majority of texts have similar sizes, which aids in model generalization. In contrast, the Hugging Face dataset's complexity with a wider variety of text lengths and some extremely long examples (exceed 800,000 characters), required more sophisticated preprocessing or tuning to achieve optimal results. For example, techniques like stop-word removal or keyword extraction can reduce the number of tokens in a long text by focusing only on the most informative words or sentences. As well, a sliding window approach that breaks the text into overlapping segments can help capture dependencies across different parts of the text while preserving the context. Furthermore, the Hugging Face dataset (100,000 rows) is significantly larger and contains a more diverse range of AI and human-written content. This likely contributed to the lower performance metrics, especially with the more complex and lengthy essays.
- **Optimization methods:** Manual tuning provided a good baseline but tended to plateau at suboptimal hyperparameter settings. While

it generally worked well for smaller datasets like Kaggle with minimal hyperparameter adjustments, it was less scalable for larger datasets with more complex requirements. In contrast, applying RS allowed BART to significantly surpass manual tuning on the Kaggle dataset. This method identified an optimal combination of learning rate and batch size, achieving high accuracy and Macro F1-score values. By randomly selecting values within specified ranges, RS had a greater chance of finding better hyperparameter combinations compared to manual tuning. Although exhaustive, the GS optimization significantly improved the BART performance on the Hugging Face dataset. Unlike RS, which selects random values, GS ensures that every possible combination of parameters is evaluated. As a result, GS has identified the need for a larger batch size on the Hugging Face dataset, which not have been explored in manual tuning. This structured approach thoroughly tested batch sizes and learning rates, had led to optimal hyperparameters and excellent performance. On the other hand, BO proved to be excellent in yielding strong performance for BART on the Hugging Face dataset. This intelligent exploration method of the hyperparameter space allowed it to effectively balance precision and recall, outperforming both manual tuning and other automated optimization methods. In fact, BO builds a probabilistic model of the function that maps hyperparameters to performance metrics such as accuracy or F1-score (Frazier, 2018). Each time a hyperparameter combination is evaluated, BO records the resulting performance and updates its probabilistic model. Using these previous evaluation results, BO predicts which new hyperparameter combinations are most likely to improve the performance, allowing it to focus on the most promising options in the hyperparameter space (Frazier, 2018). Unlike methods that rely on random sampling or exhaustive searching, BO strategically balances exploring new hyperparameter combinations with refining the promising combinations that have already performed well (Shahriari, Swersky, Wang, Adams, & de Freitas, 2016). This method reduces unnecessary evaluations, enabling faster convergence to optimal values (Shahriari et al., 2016). This explains the choice of a higher learning rate, which allowed the BART model to learn more quickly and effectively on the challenging Hugging Face dataset.

- **Learning rate impact:** The increase in learning rate during manual tuning progressively improved the performance of the models. Additionally, a learning rate of 5e-5 consistently performed well across all models and both datasets. However, BO found that increasing the learning rate to 7e-5 for BART produced even better results on the Hugging Face dataset.
- **LLMs architectures:** BART consistently outperforms GPT-2 and RoBERTa, particularly on the larger Hugging Face dataset. This can be due to BART's bidirectional encoder and autoregressive decoder, which allow it to capture more complex dependencies and nuances in the text (Lewis et al., 2020). The bidirectional encoder in BART reads the input context from both directions, enabling it to understand the text comprehensively. Meanwhile, its autoregressive

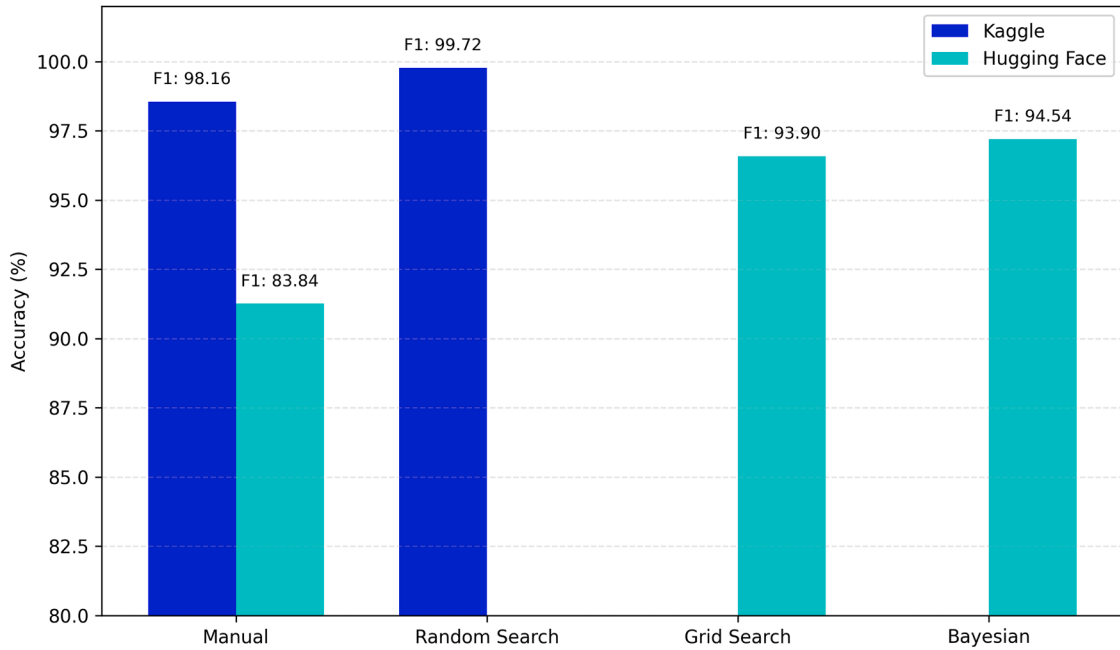


Fig. 5. The performance of our BART model across diverse optimization methods on the Kaggle and Hugging Face datasets.

decoder generates responses that maintain the sequence continuity. This combination of interpretive and generative capabilities makes BART particularly well-suited for detecting subtle patterns and inconsistencies typical of AI-generated text (Lavecchia, 2024). As a result, it has been more robust in distinguishing between AI-generated and human-written content. In contrast, GPT-2 is purely an autoregressive model, which may limit its ability to handle the diversity of text in larger datasets. GPT-2 generates each token based only on preceding tokens, from left to right, producing fluent and sequentially coherent text (Alec et al., 2019; Babaei & Giudici, 2024). However, it lacks the bidirectional view needed to fully capture nuanced contextual relationships. On the other hand, RoBERTa's bidirectional encoder is highly effective for understanding tasks, as it benefits from analyzing the context of each token on both sides (Liu et al., 2019; Rodrawangpai & Daungjaiboon, 2022). Nevertheless, its encoder-only design constrains its ability in generative understanding compared to BART.

Overall, BART clearly demonstrated its capability as the best-performing model across both datasets. Fig. 5 effectively shows the impact of optimization methods on the performance of our BART model on both Kaggle and Hugging Face datasets. Regarding optimization methods, BO provided the most efficient path to fine-tune hyperparameters, especially for large and complex datasets like Hugging Face. Additionally, RS has been also effective, particularly for BART on Kaggle dataset, achieving high and notable results. While manual tuning is useful for initial exploration, it was outperformed by more sophisticated optimization methods like RS, GS, and BO.

On the other hand, our BART-CNN model demonstrated superior consistency and robustness across all dataset configurations, outperforming the baseline BART model in both balanced accuracy and Macro F1-score. As Table 4 illustrates, BART-CNN achieved stable and high-performance results on the linguistically complex Hugging Face dataset, attaining a Macro F1-score of 98.10 and an accuracy of 98.08 under imbalanced data splits. Notably, the model maintained nearly identical performance under balanced distributions (accuracy of 98.09, and Macro F1-score of 98.08), despite the dataset's longer input sequences, higher lexical diversity, and greater syntactic variability. This performance improvement stems from our key architectural innovation in

Table 4

The performance of BART-CNN across dataset configurations (3-fold cross-validation averages).

Dataset	Split type (support ratio)	Accuracy	Macro F1-score (Avg)
Hugging Face	Imbalanced (74 % Human, 26 % AI)	98.08	98.10
	Balanced (50 % Human, 50 % AI)	98.09	98.08
Kaggle	Imbalanced (60 % Human, 40 % AI)	99.78	99.78

Fig. 2, which integrates a LayerNorm-stabilized BART encoder with a 1D CNN layer. The CNN layer captures fine-grained local token dependencies, while adaptive pooling and L2-regularized classification further enhance feature discrimination and generalization. When retrained on the Kaggle dataset using the optimal hyperparameters identified via BART-RS (Random Search), BART-CNN matched the best-reported BART-RS accuracy (99.78) while surpassing its Macro F1-score (99.78 vs. 99.73). This demonstrates that BART-CNN not only retains the benefits of hyperparameters optimization but also enhances discriminative performance through its hybrid architecture.

The BART-CNN's consistency is visually confirmed in Fig. 6, which compares the Macro F1-scores across all datasets configurations. The plot highlights our model's robustness through two key observations: (1) minimal dispersion between fold results (connected circles) for both Hugging Face configurations (HF Imbalanced/Balanced), indicating stability across data splits, and (2) near-perfect convergence of folds (Macro F1-score reached 0.9998) on the Kaggle dataset, where all folds approached the theoretical maximum. This stability is further supported by the fold-wise validation accuracies shown in Fig. 7. Remarkably, the accuracies remained tightly grouped: 97.90%–98.24% for HF Imbalanced, 97.95%–98.22% for HF Balanced, and 99.75%–99.80% for the Kaggle dataset, with training-validation gaps consistently below 2%. More critically, BART-CNN substantially outperformed all BART baselines on the challenging Hugging Face dataset, where Bayesian-optimized BART attained only a 94.54 Macro F1-score. This performance gap underscores BART-CNN's stronger generalization capability in complex textual environments, where conventional transformer architectures struggle with local feature extraction.

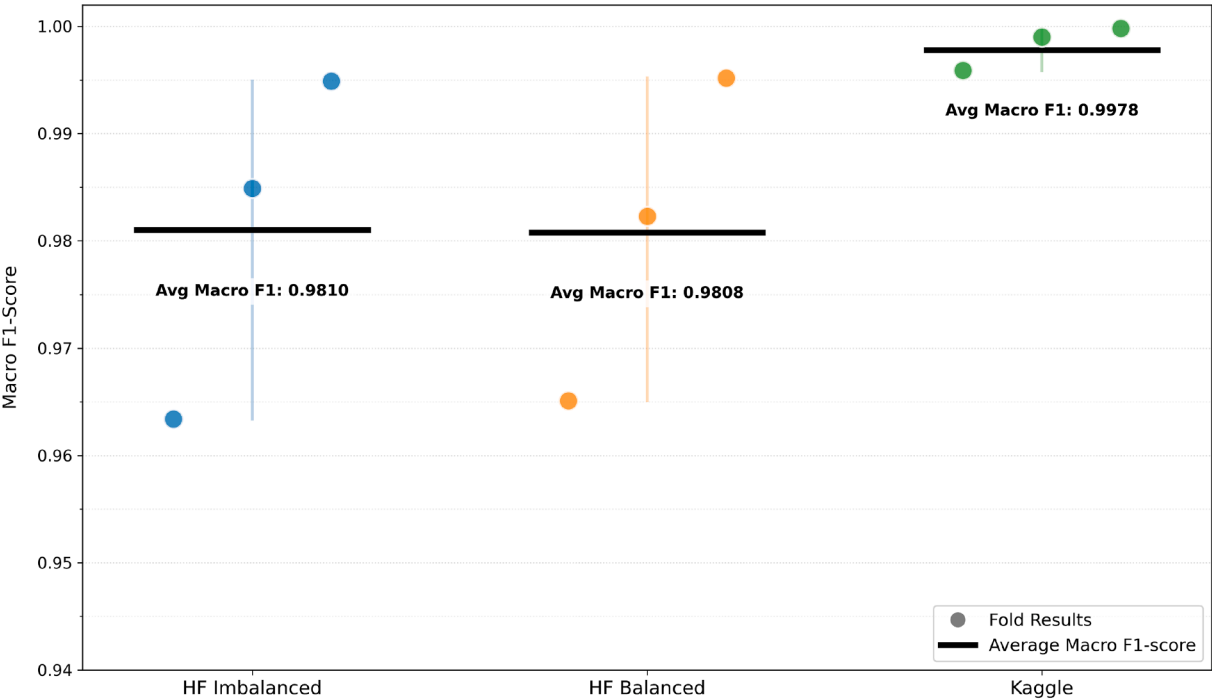


Fig. 6. Comparison of Macro F1-scores on Kaggle and Hugging Face datasets.

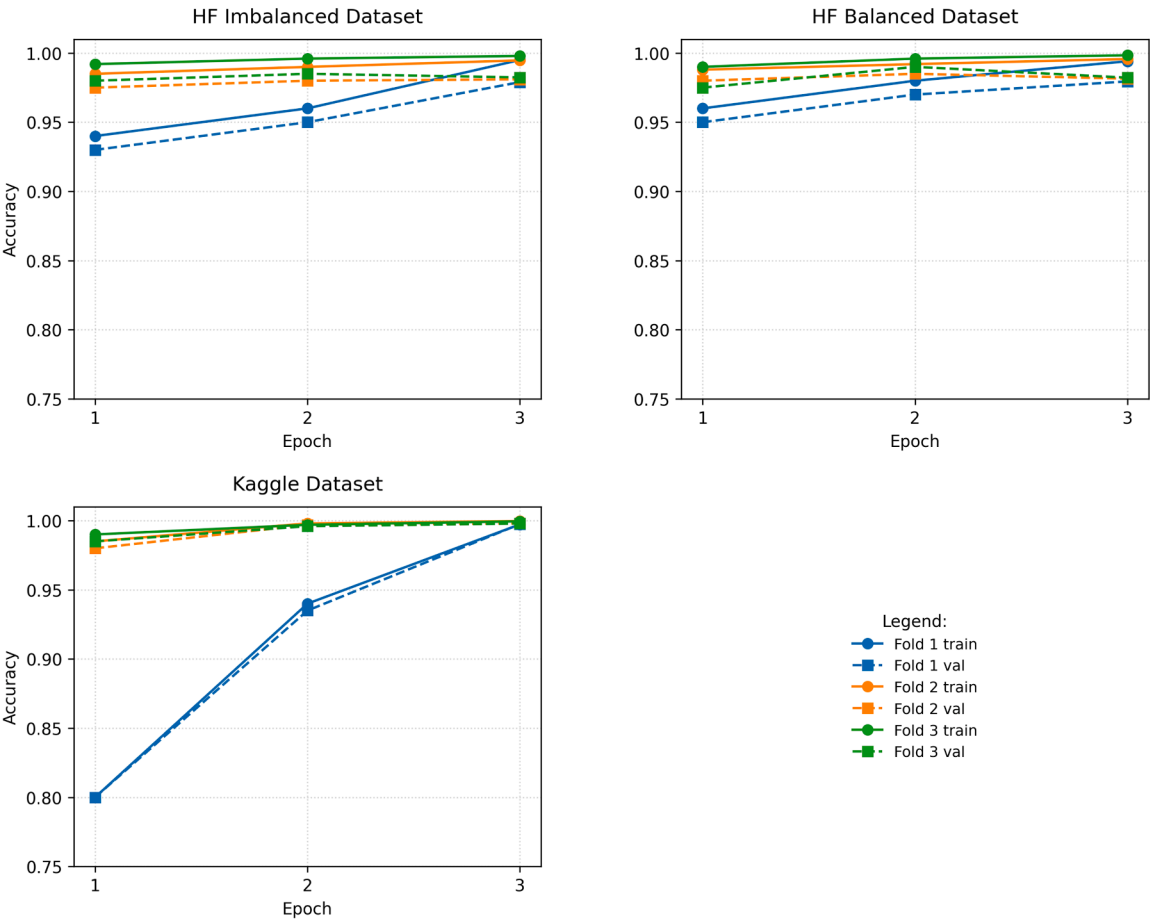


Fig. 7. The training and validation accuracy curves of BART-CNN model on Kaggle and Hugging Face datasets.

Table 5

The comparison of the accuracy and Macro F1-score of our best BART-CNN model with the baseline.

Model	Dataset	Size (H:AI)	Acc. (%)	F1-score (%)
Baseline models				
AutoEncoder (Corizzo & Leal-Arenas, 2023)	USE	335	–	95.53
ExtraTreeClassifier (Sable et al., 2024)	ChatGPT Paraph.	420K (210K:210K)	80.1	–
AdaBoostClassifier (Kartelj et al., 2025)	Abstract Paraph.	304 (152:152)	97.2	–
Our BART-CNN				
BART-CNN	USE (5-Fold)	1,957 (978:979)	99.74	99.74
BART-CNN	ChatGPT Paraph. (3-Fold)	100K (50K:50K)	100.00	100.00
BART-CNN	Abstract Paraph. (5-Fold)	304 (152:152)	97.38	98.36

Actually, the success of BART-CNN lies in its hybrid design, which synergizes the global contextual awareness of transformers with the localized pattern recognition of CNNs. While hyperparameter optimization (e.g., in BART-RS) can maximize baseline performance on simpler datasets, our model's structural enhancements address fundamental limitations of pure transformer-based approaches, particularly their tendency to overlook sub-token-level discriminative features. This explains BART-CNN's consistent gains across diverse datasets, establishing it as a more versatile and reliable solution for real-world text classification tasks. As shown in Table 5, our BART-CNN extended its significant performance on other benchmark datasets and outperformed different baseline models in identifying AI-generated and human-writing text.

These results collectively explain BART-CNN's consistent gains across diverse datasets, with different domain sources and writing styles and lengths. On the USE dataset, BART-CNN achieved near-perfect accuracy and Macro F1-score of 99.74 % each under 5-fold cross-validation, surpassing the AutoEncoder (Corizzo & Leal-Arenas, 2023) baseline which achieved a F1-score of 95.53 % by a substantial margin. Moreover, on the larger ChatGPT Paraphrasing corpus (100K samples), BART-CNN achieved flawless 100 % accuracy and Macro F1-score, and perfectly outperformed the ExtraTreeClassifier (Sable et al., 2024) accuracy of 80.1 %. Even on the smaller Abstract Paraphrasing dataset, BART-CNN edged out the AdaBoostClassifier (Kartelj et al., 2025) (97.2 % vs. 97.38 % accuracy), while also reporting a higher Macro F1-score (98.36 %), highlighting its stability across data scales. Three key factors explain BART-CNN's superiority over baseline models. First, its hybrid architecture leverages BART's contextual understanding alongside CNN's local pattern detection, enabling it to capture both syntactic anomalies (e.g., repetitive phrasing in AI text) and semantic coherence (human-like flow). In contrast, the AutoEncoder's reconstruction-based approach struggles with subtle linguistic artifacts, while tree-based methods (ExtraTree, AdaBoost) lack deep sequential modeling. Second, BART-CNN's adaptive pooling and LayerNorm stabilization mitigate overfitting, which is critical for generalizing across datasets of varying sizes, as evidenced by its stable performance on both the 304-sample Abstract set and the 100K-sample ChatGPT corpus. Third, unlike bag-of-words or statistical classifiers, BART-CNN's token-level and CNN filters explicitly model n-gram discontinuities common in AI-generated text, such as unnatural topic transitions or over-reliance on template-like structures. Actually, the results illustrate that BART-CNN not only demonstrated strong performance across evaluation metrics, but also excelled at handling paraphrased and abstract text with little word overlap, and performed consistently across different English proficiency levels. This suggests that it likely captures deeper writing patterns rather than relying on surface-level features.

However, despite its strong performance, our BART-CNN may exhibit some limitations that warrant discussion. First, our model currently operates as a strict binary classifier (AI vs. human), while real-world applications might benefit from soft classification outputs that provide confidence scores or percentage-based likelihood estimates. This would be particularly valuable for borderline cases where text exhibits mixed human/AI characteristics. Second, while our model maintains robust performance across text lengths, its implementation employs distinct

token limits optimized for different dataset characteristics. For instance, it used a conservative 300-token window for large-scale datasets (e.g., ChatGPT Paraphrasing with 100K samples) to ensure computational efficiency during bulk processing, and an extended 1024-token limit for the Abstract Paraphrasing dataset to preserve nuanced academic phrasing. This design choice, while effective for our experimental scope, introduces two considerations: (1) academic samples exceeding 1024 tokens must be truncated, which may result in the loss of important contextual information, and (2) the fixed 300-token approach for large datasets may theoretically omit distant linguistic patterns in exceptionally long-form human writing. Third, our current implementation doesn't explicitly model temporal evolution of writing styles, which could be crucial for detecting AI-generated text that adapts to mimic human writing patterns over time. These limitations point to valuable future research directions. Our BART-CNN model could be extended to provide probabilistic outputs rather than binary decisions, enabling more nuanced detection scenarios. Additionally, overcoming our current computational constraints could enable better handling of variable-length documents while preserving our model's efficiency advantages. Furthermore, while our BART-CNN demonstrates strong performance on English text, its architecture could be adapted for multilingual detection by incorporating language-specific tokenization and pretraining. Collectively, these developments would further enhance our model's practical utility in real-world content moderation and authenticity verification systems.

5. Use case scenarios

Our BART-CNN model's robust performance and architectural advantages make it as a valuable tool for several real-world applications. In academic integrity enforcement, the model could be deployed by journal editors and peer-review systems to screen manuscript submissions for AI-generated text, particularly in identifying paraphrased content that evades traditional plagiarism detectors (Chaka, 2023; Kulka-rni et al., 2021). While current token limits require preprocessing for book-length texts, our model's proven accuracy on academic abstracts (98.36 % Macro F1-score) suggests immediate utility for conference submission systems or preprint repositories combating papermill-generated research (Elkhatat, Elsaid, & Almeer, 2023; Michel-Villarreal, Vilalta-Perdomo, Salinas-Navarro, Thierry-Aguilera, & Gerardou, 2023; Neumann, Rauschenberger, & Schön, 2023). For online content moderation, its efficiency at scale makes it suitable for social media platforms needing to flag synthetic disinformation campaigns while minimizing false positives on human-authored posts (Gulzar Hussain et al., 2025; Saeidnia et al., 2025; Zhou, Zhang, Luo, Parker, & De Choudhury, 2023). Besides, educational institutions could integrate our model into writing assistance tools to help instructors discern AI-generated student work, especially given its balanced performance on short-form responses (e.g., essay questions) and longer compositions (Ardito, 2024; Cingillioglu, 2023; Dalalah & Dalalah, 2023). In effect, our BART-CNN model's nuanced detection capability also addresses emerging challenges in legal and financial domains, where distinguishing between human- and AI-drafted contracts or reports carries ethical and compliance implications (Autto, Haapio, & Nuottila, 2024; Giampieri, 2024; Wang, 2024). These

applications highlight our BART-CNN's practical value across sectors where AI-text provenance is critical, though its adoption would benefit from the aforementioned improvements in variable-length processing and probabilistic outputs.

6. Ethical considerations and practical safeguards

While our BART-CNN offers powerful capabilities for AI-generated text detection, its deployment raises ethical questions that demand proactive mitigation through both technical and operational safeguards. First, false positives (misclassifying human-written text as AI-generated) pose a significant ethical risk, potentially leading to unjust penalties for students, researchers, and content creators (Dalalah & Dalalah, 2023; Elkhatat et al., 2023). To mitigate this, we recommend that all positive classifications from our binary BART-CNN model undergo mandatory human review. Additionally, we propose a multi-stage verification pipeline that integrates our model's outputs with document metadata analysis to enhance decision reliability. Looking forward, our model architecture could be adapted to produce probabilistic confidence scores (e.g., an 85% likelihood of AI authorship) by exposing the softmax layer outputs, replacing rigid binary judgments. This probabilistic scoring would enable institutions to implement threshold-based policies, for instance, flagging only cases below a 95% confidence threshold for mandatory human review. In addition, to safeguard against erroneous or biased outcomes, transparent appeal processes and mandatory human-in-the-loop oversight for high-stakes decisions (such as allegations of academic misconduct) should be institutionalized (Carobene, Padoan, Cabitza, Banfi, & Plebani, 2023; Langer, Baum, & Schlicker, 2024). These safeguards collectively help ensure that our BART-CNN model's use remains fair, accountable, and aligned with ethical standards. Second, while the effectiveness of our model opens the door to surveillance-oriented applications, such as monitoring content on social media platforms, we emphasize the importance of robust governance frameworks (Villegas-Ch & García-Ortiz, 2023; Wen, Shukla, & Katt, 2024). These should mandate opt-in policies, on-premise deployment, and clearly defined purpose limitations to ensure that detection capabilities do not come at the expense of individual privacy rights (Kafali, Preuveneers, Semertzidis, & Daras, 2024). Moreover, to mitigate the risk of adversarial misuse where malicious actors probe decision boundaries to evade detection, our model leverages a hybrid CNN-transformer architecture, which inherently resists overfitting and pattern memorization. Additionally, restricting API access to authorized users helps prevent large-scale misuse or automated probing of the model's decision boundaries (Villegas-Ch & García-Ortiz, 2023; Wen et al., 2024). Furthermore, although our evaluation across five benchmark datasets revealed less than a 2% difference in Macro F1-score between demographic groups, concerns about cultural and linguistic biases in the training data remain valid (Liang, Yuksekgonul, Mao, Wu, & Zou, 2023). To address this, we recommend ongoing dynamic fairness audits throughout deployment, supported by cultural-linguistic review boards tasked with monitoring potential disproportionate impacts, particularly on non-native English writers. These procedural safeguards align with our proposed technical enhancements for multilingual adaptation, as outlined in the BART-CNN model limitations. Collectively, these safeguards: confidence thresholds, human review tiers, access controls, and bias monitoring, ensure that our BART-CNN model's detection capabilities align with ethical AI principles while preserving its performance advantages.

7. Conclusion and future work

In this paper, we fine-tuned three LLMs, including BART, GPT-2, and RoBERTa, for the task of detecting AI-generated essays using extensive datasets from the Hugging Face and Kaggle platforms. In addition, we conducted an in-depth analysis of how automated optimization methods impact our LLMs accuracies and efficiencies. Our experiments demonstrated that BO significantly enhanced the performance

of the BART model by identifying an optimal and balanced combination of hyperparameters. Furthermore, both RS and GS provided substantial improvements over manual tuning, particularly in detecting AI-generated and human-writing contents. Building on these findings, we developed a hybrid BART-CNN model that effectively integrates the contextual understanding of LLMs with the local feature extraction capabilities of CNNs. Our proposed BART-CNN model achieved remarkable performance across multiple datasets, attaining Macro F1-scores of 99.78% and 98.10% on the Kaggle and Hugging Face datasets, respectively. It also demonstrated strong generalization capabilities on benchmark datasets, achieving accuracies of 99.74% on the USE dataset, 100% on ChatGPT Paraphrased essays, and 97.38% on Abstract Paraphrased texts. These results confirm the model's robustness and effectiveness in identifying human-writing and AI-generated text. For future work, additional datasets may be explored to further improve the generalizability of our models across languages and writing styles. Overall, our paper contributes to the development of advanced AI-driven plagiarism detection tools in the era of generative language models, thereby supporting more authentic and reliable assessment practices in academic contexts.

CRedit authorship contribution statement

Mohammed Qorich: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Writing – review & editing, Visualization; **Rajae El Ouazzani:** Supervision, Project administration, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Availability of data and materials

All results presented in this paper are reproducible. All accompanying resources to this work can be available upon reasonable request to the corresponding author. The datasets are publicly available, and the links are shared in The datasets preparation Section (3.1).

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